

Mathematics for Machine Learning

Vazgen Mikayelyan



Signal Processing

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 - A signal is mathematically just a function.
- What is a signal processing?
 - Convert one signal to another.
 - Information extraction and interpretation.

Fourier Transform

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Definition

Let $f : \mathbb{R} \rightarrow \mathbb{C}$. The Fourier transform of f is defined as follows

$$\mathcal{F}[f](x) = \hat{f}(x) = \int_{-\infty}^{+\infty} f(t) e^{-2\pi i x t} dt.$$

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For all $p \in [1, \infty)$ denote by L^p the set of all functions f , such that

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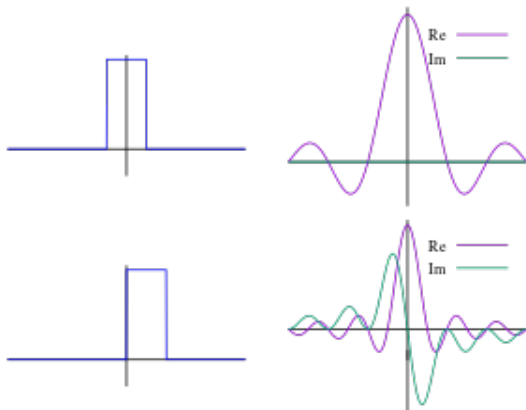
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Theorem

If $f \in L^1$, then the Fourier transform of f exists.

Fourier Transform



Properties

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Plancherel's Theorem

If $f \in L^1 \cap L^2$, then $\hat{f} \in L^2$ and

$$\int_{-\infty}^{+\infty} |\hat{f}(x)|^2 dx = \int_{-\infty}^{+\infty} |f(t)|^2 dt.$$

Convolution

Definition

Convolution of the functions $f, g : \mathbb{R} \rightarrow \mathbb{R}$, $f, g \in L^1$, is defined as the integral of the product of the two functions after one is reversed and shifted:

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Properties

- 1 $f * g = g * f$,
- 2 $(f * g) * h = f * (g * h)$,

Young's Convolution Inequality

If $f \in L^p$, $g \in L^q$ and $\frac{1}{p} + \frac{1}{q} = 1 + \frac{1}{r}$, with $1 \leq p$ and $1 \leq q \leq r < \infty$, then $\|f * g\|_r \leq \|f\|_p \|g\|_q$.

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Convolution Theorem

If $f, g \in L^1$, then $\mathcal{F}[f * g] = \mathcal{F}[f] \mathcal{F}[g]$.

Inverse Fourier Transform

Definition

Let $g : \mathbb{R} \rightarrow \mathbb{C}$. The inverse Fourier transform of f is defined as follows

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Theorem

If $f, \hat{f} \in L^1$ then $\mathcal{F}^{-1}[\hat{f}] = f$, almost everywhere on $\mathbb{R}$.

Digital Signal Processing

- What does digital mean?

Digital Signal Processing

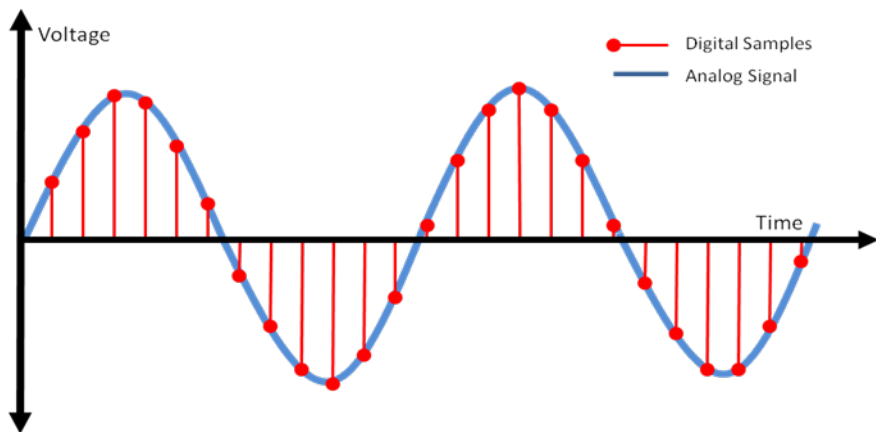
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Digital Signal Processing

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- Can be discrete by nature.

Example of Sampling



- **Audio recording:**

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- **Photos:** the analog scene of light is sampled using a CCD array and stored as a two-dimensional discrete-space signal
- **Ratings:** for books (Goodreads), movies (Netflix), vacation rentals (Air bnb) are stored using the integers 0-5

- Speaker Identification

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- Speech to Text

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- Source Separation

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- Discrete signals:

$$x[n] : \mathbb{Z} \rightarrow \mathbb{R} \text{ or } x[n] : \mathbb{Z} \rightarrow \mathbb{C}.$$

- Delta Signal:

$$\delta_n = \begin{cases} 1, & \text{if } n = 0, \\ 0, & \text{if } n \neq 0 \end{cases}$$

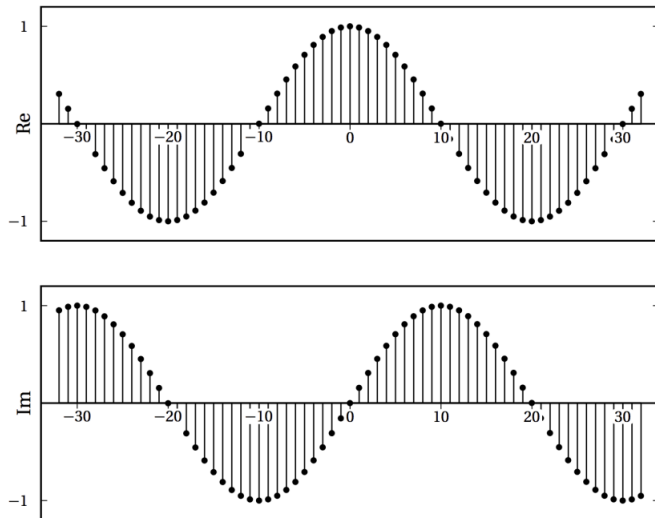
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- Complex Exponential Signal:

$$x[n] = e^{i(\omega_0 n + \varphi)}$$

Complex Exponential Signal



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- ω_0 is the frequency in radians per sample.
- Frequency in cycles per sample $f = \frac{\omega_0}{2\pi}$.

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A discrete time signal is periodic if there is an $n_0 \in \mathbb{Z}$ such that $x[n] = x[n + n_0]$, for all $n \in \mathbb{Z}$.

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Theorem

A discrete-domain or discrete-time sinusoid is periodic if and only if its frequency ω_0 is π times a rational number, that is

$$\omega_0 = \frac{M}{N}\pi, M, N \in \mathbb{Z}.$$

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Energy of a discrete signal $x[n]$, $n \in \mathbb{Z}$ is defined as follows

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$$P_x = \lim_{N \rightarrow \infty} \frac{1}{2N+1} \sum_{n=-N}^N |x[n]|^2.$$

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Let us suppose that any analog signal can be represented as a sum of sinusoids of different amplitudes, frequencies and phases, i.e.

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All signals such that speech and video, lend themselves to such a representation over any short time segment. **For speech the maximum frequency is approximately 3400 Hz.**

Sampling Theorem

Nyquist–Shannon

If the highest frequency contained in an analog signal $x_a(t)$ is F_{max} and the signal is sampled at a rate $F_s > 2F_{max}$, then $x_a(t)$ can be exactly recovered from its sample values using the interpolation function

$$g(t) = \frac{\sin(2\pi F_{max}t)}{2\pi F_{max}t}.$$

Thus $x_a(t)$ can be expressed as

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The sampling rate $F_N = 2F_{max}$ is called the Nyquist rate.

Fourier Series

Definition

The function $T : \mathbb{R} \rightarrow \mathbb{R}$ is called a trigonometric polynomial of degree n , if there exist sequences of real numbers $\{a_k\}_{k=0}^n, \{b_k\}_{k=1}^n$ such that $a_n^2 + b_n^2 > 0$ and

$$T(x) = a_0 + \sum_{k=1}^n (a_k \cos kx + b_k \sin kx).$$

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